

Fisher-Orthogonal Parameter Manifolds: Enforcing Hard Constraints in Hypernetwork-based Continual Learning

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1 Project Description

This project integrates **Fisher-Orthogonal Projected Natural Gradient Descent (FOPNG)** (Garg et al., 2026) within a Hypernetwork architecture to address **catastrophic forgetting** (McCloskey & Cohen, 1989). By introducing a "hard" geometric constraint to the generator's optimization process, I aim to preserve the integrity of previously learned task representations. In standard **Hypernetworks** (Ha et al., 2016), the generator often overwrites parameter manifolds essential for prior tasks. The proposed modification calculates the **Fisher Information Matrix (FIM)** (Amari, 1998) to identify load-bearing parameters and projects subsequent gradients into their null-space.

This approach ensures weight-generation logic remains invariant while acquiring new knowledge, treating consolidation as a manifold-squeezing problem rather than a simple optimization task. Beyond evaluating task retention, I will investigate the **Manifold Capacity** of the Hypernetwork to determine exactly how many tasks can be 'packed' before it reaches saturation. I will verify the optimality of these orthogonal separations by comparing my results against the soft-regularization approach of **Elastic Weight Consolidation (EWC)** (Kirkpatrick et al., 2017). Finally, I will explore a modified FOPNG variant with flexible constraints to test if controlled subspace overlap improves efficiency without sacrificing memory.

2 Motivation

Addressing **catastrophic forgetting** (McCloskey & Cohen, 1989) is a critical challenge because standard neural architectures prioritize immediate optimization over long-term stability. Sequential learning has vital security implications: as noted in the **FOPNG framework** (Garg et al., 2026), the ability to learn without re-accessing past training data is essential for privacy-sensitive fields like healthcare or personalized edge computing. However, without mathematically sound memory management, new training sessions can silently corrupt verified, safety-critical logic when the original data is unavailable for re-verification.

This project extends original FOPNG research by transitioning from standard MLP/CNN architectures to a **Hypernetwork** framework (von Oswald et al., 2020; De La Fuente et al., 2025). This higher-level abstraction protects the **weight-generation manifold** itself—preserving the model's underlying "logic" for creating features rather than just the features themselves.

Scientifically, this research explores whether shifting from "soft" heuristic penalties (Kirkpatrick et al., 2017) to the mathematically rigorous "hard" constraints of FOPNG (Garg et al., 2026) provides a more reliable foundation for memory. Beyond absolute retention, a critical and under-explored question is the **packing efficiency** of these task representations. Following the geometric theory of perceptual manifolds (Chung et al., 2018), this study seeks to investigate **Information Density**; quantifying how proficiently the FOPNG algorithm occupies the Hypernetwork's parameter space before reaching a point of **Saturation**.

From an engineering perspective, this work moves toward resource-conscious architectures that intelligently allocate storage. This is a necessary step for building AI that is not only more capable but also easier to deploy on constrained devices where memory and data privacy are the primary bottlenecks.

3 Research Question

To what extent does the integration of Fisher-Orthogonal Projected Natural Gradients (FOPNG) within a Hypernetwork architecture optimize the trade-off between absolute task retention and manifold capacity in sequential learning?

4 Background

4.1 Catastrophic Forgetting and the Stability-Plasticity Dilemma

Central to the challenge of sequential learning is the **stability-plasticity dilemma** (Grossberg, 1980). This dilemma posits that a learning system must resolve two conflicting requirements: it must be **'plastic'** enough to integrate novel patterns, yet **'stable'** enough to prevent the erosion of established memories. In artificial neural networks, the high plasticity required for backpropagation often leads to **catastrophic forgetting** (McCloskey & Cohen, 1989), where new gradient updates move the model parameters away from the manifolds representing prior tasks.

4.2 Hypernetworks as Weight Generators

Hypernetworks are meta-architectures designed to generate the **parameters (weights)** for a secondary **target network** (Ha et al., 2016). In this paradigm, the model learns a mapping from a task embedding directly to the target model’s weight space.

Integrating Hypernetworks into **Continual Learning (CL)** addresses the **stability-plasticity dilemma** by shifting optimization to a meta-parameter space. The network generates task-specific weights using a task identifier, effectively partitioning knowledge across different objectives (von Oswald et al., 2020). By protecting the hypernetwork weights rather than the target network’s fluctuating parameters, the system achieves **high plasticity** for new tasks while ensuring the **stability** of prior solutions. Current research further optimizes this via **partial hypernetworks** to minimize **computational overhead** (Hemati et al., 2023).

4.3 Information Geometry and the Fisher Information Matrix

Information Geometry provides a framework for treating parameter space as a **Riemannian manifold**. Within this space, the **Fisher Information Matrix (FIM)** serves as a **metric tensor** defining local curvature relative to the model’s output distribution (Amari, 1998). Formally, for a model parameterized by θ , the FIM ($\mathcal{F}$) represents the expected second-order derivative of the log-likelihood:

$$\mathcal{F} = \mathbb{E}_{p(x,y|\theta)} [\nabla_{\theta} \log p(y|x, \theta) \nabla_{\theta} \log p(y|x, \theta)^T] \quad (1)$$

By accounting for this curvature, the **Natural Gradient** enables optimization that is invariant to parameterization, following the steepest descent on the manifold rather than in Euclidean space. In this project, the FIM identifies the **“load-bearing” parameters** within the **Hypernetwork**. By quantifying how sensitive task-weights are to infinitesimal changes in meta-

parameters, the FIM reveals which dimensions of the manifold carry the specific knowledge required for previously learned tasks.

4.4 Regularization vs. Geometric Constraints: From EWC to FOPNG

The evolution of CL methods is characterized by a shift from heuristic penalties to rigid geometric constraints. Early approaches, such as **Elastic Weight Consolidation (EWC)** (Kirkpatrick et al., 2017), utilize the FIM to implement “soft” regularization via a quadratic penalty. However, soft regularization often suffers from **drift**, where cumulative updates for new tasks erode prior knowledge.

While standard **Euclidean orthogonal projections** can isolate task updates, they ignore the information-theoretic structure of the parameter manifold. This project investigates the **Fisher-Orthogonal Projected Natural Gradient (FOPNG)** (Garg et al., 2026), which enforces a “hard” geometric constraint by projecting new gradients into the **null-space** of the aggregate FIM. By respecting the Riemannian metric of the network, this ensures Hypernetwork updates remain **mathematically invariant** relative to prior task logic.

5 Experimental Design and Dataset Description

This study evaluates task retention and manifold capacity using established CL benchmarks, ordered by increasing complexity to isolate the effects of geometric constraints.

5.1 Benchmarks and Datasets

The evaluation will proceed in three stages:

- **Stage 1 (Validation): Split-MNIST (LeCun et al., 1998).** Five sequential digit pairs to test basic task separation and verify the FOPNG-Hypernetwork integration.
- **Stage 2 (Capacity Analysis): Permuted-MNIST (Goodfellow et al., 2013).** Ten sequential tasks using random pixel-permutations of the original MNIST digits.
- **Stage 3 (High-Dimensional Scaling): Split-CIFAR-10/100 (Krizhevsky, 2009) and Mini-ImageNet (Deng et al., 2009).** Complex visual datasets used to evaluate how the Fisher-Orthogonal projection scales to high-dimensional manifolds.

5.2 Comparative Architecture Design

The experiment utilizes three primary configurations:

1. **Baseline (Soft Regularization):** A standard Hypernetwork using output-fixing regularization (von Oswald et al., 2020).
2. **Experimental A (Hard Constraint):** A Hypernetwork integrated with the **Pre-Fisher (Truncated Gradient)** algorithm from the FOPNG framework (Garg et al., 2026) for absolute task retention.
3. **Experimental B (Flexible Constraint):** A modified FOPNG variant with a relaxation hyperparameter to test if controlled subspace overlap improves information density. This will be conducted after the Hard Constraint baseline is established.

5.3 Evaluation Metrics

Beyond classification accuracy and Backward Transfer (BWT), we track:

- **Gradient Norm Ratio (ρ):** $\rho = \frac{\|g_{proj}\|}{\|g\|}$, indicating remaining capacity and the **Saturation Point** where $\rho \rightarrow 0$.
- **Effective FIM Rank:** A spectral analysis of the cumulative FIM to quantify manifold occupation proficiency.
- **Fisher Overlap:** Defined as $1 - d^2$, where d is the Fréchet distance between task Fisher matrices. A value of zero indicates non-overlapping parameter sensitivities, while a value of one indicates tasks depend on identical parameter profiles.

6 Instrumentation and Methodology

The research is computational, utilizing **Python 3.10** and **PyTorch**. The architecture is a **non-chunked Hypernetwork**, which simplifies the parameter manifold for transparent spectral analysis.

6.1 Algorithmic Framework

The implementation synthesizes logic from the *HyperCL* (von Oswald et al., 2020) and *FOPNG* (Garg et al., 2026) repositories. I will prioritize the implementation of the **Pre-Fisher algorithm**, which utilizes a diagonal approximation of the Fisher Information Matrix to ensure computational feasibility. **NumPy/SciPy** handles the Singular Value Decomposition (SVD) for null-space projection. Computation will be performed on CUDA-enabled GPUs to manage the iterative linear algebra operations.

6.2 Statistical and Geometric Analysis

The project milestones are ordered by technical necessity:

- **Baseline Validation:** Results are reported as mean accuracy and standard error across **5 unique random seeds** for the hard-constraint FOPNG model.
- **Spectral Analysis:** Qualitative analysis of the **Fisher Information Trace** to verify if the "Geometric Shield" correctly isolates and protects load-bearing parameters.
- **Ablation Study:** Inferential analysis on the FOPNG **relaxation hyperparameter** (Experimental B) to determine the optimal trade-off between strict orthogonality and manifold efficiency.

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